GITAM Deemed to be University

Artificial Intelligence and Machine Learning | Course Code: AI201

Artificial Intelligence & Search Strategies - AI Problem Solving, Heuristic

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Overview: Completeness and optimality proofs of A*, admissible heuristics, Alpha-Beta pruning,

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Intelligent Agents & State Space Search
Agent architectures, PEAS definitions, Breadth-First, Uniform Cost, Depth-First, and Iterative Deepe

Unit 2: Informed Heuristic Search
Greedy best-first search, A* search algorithm, Admissible and consistent heuristics, Memory-bounded

Unit 3: Adversarial Search & Games
Game theory formulation, Minimax decision algorithm, Alpha-Beta pruning optimality, Evaluation funct

Unit 4: Constraint Satisfaction & Logic
Constraint propagation, AC-3 algorithm, Backtracking with MRV heuristic, Propositional and First-Ord

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Prove that A* search is optimal when the heuristic function hn is admissible for tree search and consi
2. Apply Alpha-Beta pruning on a specified 4-ply minimax game tree and show the branches that are pruned.
3. Solve the 4-Queens Constraint Satisfaction Problem using backtracking search and arc-consistency AC-3

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Stuart Russell, Peter Norvig - Artificial Intelligence: A Modern Approach Pearson 4th Ed
* Elaine Rich, Kevin Knight - Artificial Intelligence McGraw-Hill

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